In other words the square of the uncertainty is the average value of the operator $\mathbf{A}^2$.

5.5 CAUCHY-SCHWARZ INEQUALITY

The uncertainty principle is an inequality that says the product of the uncertainties of **A** and **B** is larger than something that involves their commutator. The basic mathematical inequality is the familiar triangle inequality. It says that in any vector space, the magnitude of one side of a triangle is less than the sum of the magnitudes of the other two sides. For real vector spaces, we derive

$$|X||Y| \geq |X \cdot Y| \tag{5.5}$$

from the triangle inequality,

$$|X| + |Y| \geq |X + Y|.$$

5.6 THE TRIANGLE INEQUALITY AND THE CAUCHY-SCHWARZ INEQUALITY

The triangle inequality is motivated, of course, by the properties of ordinary triangles, but it's actually far more general and applies to a large class of vector spaces. You can get the basic idea by looking at Fig. 5.1, where the sides of the triangle are taken to be ordinary geometric vectors in a plane. The triangle inequality is just the statement that the sum of any two sides is bigger than the third side, and the underlying idea is that the shortest path between two points is a straight line. The shortest path between point 1 and point 3 is side *Z*, and the sum of the other two sides is certainly bigger.